

Sifan Tao

Curriculum Vitae

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Education

- 2023 – Present **PhD Student in Statistics**, *University of Wisconsin-Madison*.
- 2019 – 2023 **BS in Statistics, with a second major in mathematics**, *University of Virginia*.
with distinction, Phi Beta Kappa
- Fall 2020 **UVA at Fudan: Go Local**, *School of Data Science, Fudan University*.

Publications

Journal Articles

- to appear **Sifan Tao** and Peter Chien. Group representative Latin hypercube design. *SIAM/ASA Journal of Uncertainty Quantification*, to appear.

Submitted

Ziqi Yang, **Sifan Tao**, Qianli Xing, and Fang Liu. Spatiotemporal Origin and Dynamics of Sodium Plating in Sodium-Ion Batteries.

Sifan Tao and Peter Chien. Robust Latin Hypercube Design for Computer Experiments with Potential Failure Regions.

Xiaoyu Lei*, **Sifan Tao***, Peter Chien, and Joshua Cape. Overlapping sliced Latin hypercube designs for fusing multiple computer models. *: equal contribution.

Presentations

Conference Poster Presentations

- May 2025 **Uncertainty Quantification and Machine Learning for Complex Physical Systems**, *Variance Reduction under Grouped Variables in Sobol Models*, Institute for Mathematical and Statistical Innovation, Advised by Peter Chien.

Invited Talks

- November 2021 **Ohio State Sports Analytics Association Conference 2021**, *Modeling Stochastic Soccer Game Dynamics using Potential Functions*, Ohio State University, Joint work with Nicholas Ho, Adhvaith Vijay.
- October 2021 **Carnegie Mellon Sports Analytics Conference 2021**, *An Interpretable Method of Learning Stochastic Game Dynamics*, Carnegie Mellon University, Joint work with Nicholas Ho, Adhvaith Vijay.

Invited Guest Lectures

- October 2022 **STAT4800 Advanced Sport Analytics**, *Modeling Stochastic Soccer Game Dynamics using Potential Functions*, University of Virginia, Invited by Professor Jordan Rodu.
Undergraduate course on sport analytics.

Industry Experience

- Summer 2026 **Data Science PhD Intern**, *Capital One*, Upmarket Acquisition Behavioral, McLean, VA.

Student Research Experience

- Fall 2024 – **Graduate Research Assistant**, *Leverage-based Orthogonalizing EM Method*, Advised by Peter Chien.
Present University of Wisconsin-Madison
- Fall 2021 – **Undergraduate Research Assistant**, *Core-Periphery Trading Networks*, Advised by Chaojun Wang.
Fall 2022 The Wharton School, University of Pennsylvania
- Fall 2021 – **Undergraduate Research Course**, *Bayesian Propensity Scores, Sentiment analysis of Covid-19 tweet*, Advised by Xin Tong.
Spring 2022 University of Virginia
- Fall 2021 **Undergraduate Independent Study Course**, *Causal Inference*, Advised by Jordan Rodu.
University of Virginia
- Summer 2021 **Summer Undergraduate Research Experience**, *Modeling Stochastic Soccer game Dynamics using Potential functions*, Advised by Konstantinos Pelechrinis.
Carnegie Mellon Sports Analytics Camp 2021
- March 2021 – **Undergraduate Research Assistant**, *Why Trade Over-the-counter? When Investors Want Price Discrimination*, Advised by Chaojun Wang.
April 2021 The Wharton School, University of Pennsylvania

Service

Professional Service

- Reviewer Journal of Statistical Planning and Inference

Teaching

Courses Served as Teaching Assistant at University of Wisconsin-Madison

- Fall 2026 **STAT424 Experimental Design.**
- Spring 2025 – **STAT340 Data Science Modeling II.**
Spring 2026
- Fall 2024 **STAT240 Data Science Modeling I.**
- Spring 2024 **STAT371 Introductory Applied Statistics for the Life Science.**
- Fall 2023 **STAT324 Introductory Applied Statistics for Engineers.**

Courses Served as Teaching Assistant at University of Virginia

- Fall 2021, **STAT3080 From Data to Knowledge.**
Fall 2022

Computer Skills

- Proficient R
- Intermediate Python
- Beginner Shell, Java, SAS

Language

- Bilingual Chinese, English
- Beginner Japanese